

LESSON FOCUS

Explore factoring polynomials with algebra tiles.

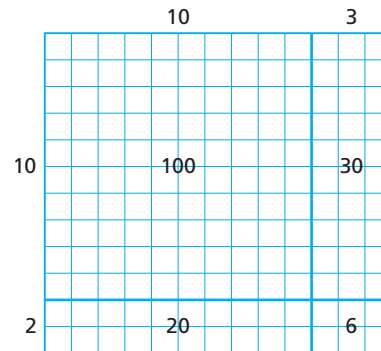


Make Connections

We can use an area model and the distributive property to illustrate the product of two 2-digit numbers.

$$\begin{aligned}
 12 \times 13 &= (10 + 2)(10 + 3) \\
 &= 10(10 + 3) + 2(10 + 3) \\
 &= 10(10) + 10(3) + 2(10) + 2(3) \\
 &= 100 + 30 + 20 + 6 \\
 &= 156
 \end{aligned}$$

How could you use an area model to identify the binomial factors of a trinomial?



Construct Understanding

TRY THIS

You will need algebra tiles. Use only positive tiles.

A. Use 1 x^2 -tile, and a number of x -tiles and 1-tiles.

- Arrange the tiles to form a rectangle. If you cannot make a rectangle, use additional x - and 1-tiles as necessary. For each rectangle, sketch the tiles and write the multiplication sentence it represents.
- Choose a different number of x -tiles and 1-tiles and repeat until you have 4 different multiplication sentences.

- B.** Use 2 or more x^2 -tiles and a number of x -tiles and 1-tiles.
- Arrange the tiles to form a rectangle. Use additional x - and 1-tiles if necessary. For each rectangle, sketch the tiles and write the corresponding multiplication sentence.
 - Repeat with different numbers of tiles until you have 4 different multiplication sentences.
- C.** Share your work with a classmate. What patterns do you see in the products and factors?

Assess Your Understanding

- Which of the following trinomials can be represented by a rectangle? Use algebra tiles to check. Sketch each rectangle.

a) $y^2 + 4y + 3$	b) $d^2 + 7d + 10$	c) $m^2 + 7m + 7$
d) $r^2 + 14r + 14$	e) $t^2 + 6t + 6$	f) $p^2 + 9p + 2$
- Which of the following trinomials can be represented by a rectangle? Use algebra tiles to check your answers. Sketch each rectangle.

a) $2s^2 + 7s + 3$	b) $3w^2 + 5w + 2$	c) $2f^2 + 3f + 2$
d) $2h^2 + 10h + 6$	e) $4n^2 + 2n + 1$	f) $6k^2 + 11k + 3$
- Suppose you must use 1 x^2 -tile and twelve 1-tiles. Which numbers of x -tiles could you use to form a rectangle?
- Suppose you must use 2 x^2 -tiles and 9 x -tiles. Which numbers of 1-tiles could you use to form a rectangle?



THE WORLD OF MATH

Math Fact: Taylor Polynomials

A certain type of polynomial is named for the English mathematician Brook Taylor, 1685–1731. Taylor polynomials are used to determine approximate values.

When you use the square root key on a scientific calculator, you are seeing Taylor polynomials in action. The square root of a number between 0 and 2 can be approximated using this polynomial:

$$\sqrt{x} \doteq 1 + \frac{1}{2}(x - 1) - \frac{1}{8}(x - 1)^2 + \frac{1}{16}(x - 1)^3 - \frac{5}{128}(x - 1)^4$$

Use the polynomial to estimate the value of $\sqrt{1.5}$.

Use a calculator to determine how many digits of your answer are correct.